

JAYANTH DASAMANTHARAO

New Brunswick, NJ — (848)313-5617 — jayanthdasamantharao@gmail.com — linkedin.com/in/djayanth

PROFESSIONAL EXPERIENCE

Data Scientist - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Designed and deployed predictive classification and regression models over 5M+ interaction records, improving engagement by 30% through structured feature engineering and model optimization.
- Conducted large-scale statistical analysis and hypothesis testing to identify behavioral trends and key drivers of user activity, informing product and targeting strategies.
- Built anomaly detection and pattern recognition models to surface deviations in usage metrics, enabling proactive issue detection and KPI stabilization.
- Designed and built production LLM-enabled applications integrating prompting, embeddings, tool use, and RAG over enterprise datasets, supporting semantic search, content understanding, cross-modal relevance.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Built and deployed classification models using XGBoost and H2O AutoML, improving prediction accuracy by 65% through advanced feature engineering and structured validation.
- Performed exploratory data analysis and large-scale SQL-based data processing to uncover performance drivers and optimize campaign targeting strategies.
- Designed repeatable experimentation workflows and performance measurement dashboards to assess model impact and business outcomes.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, and indexed lookups to efficiently clean, deduplicate, transform large datasets, reducing 50% production issues.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
ML & Statistics: End-to-end ML pipelines, model training, evaluation, monitoring, feature engineering, MLOps.
Evaluation: Golden datasets, A/B testing, human-in-the-loop review, telemetry, monitoring, regression detection.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Sagemaker, EMR), GCP, Azure, Pinecone, Chroma.
Big Data: Spark (conceptual/SQL workflows), large-scale data processing, distributed data systems.
GenAI & LLMs: Transformer-based models, LLM fine-tuning, retrieval-augmented generation, multimodal modeling.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ
Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

LLM-Powered Retrieval & Question Answering System (Production): Built and shipped LLM-powered retrieval and question answering services using embeddings, RAG, and evaluation-driven iteration, improving response quality, factual grounding, and production reliability.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>